Results Draft 2

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1 Results

1.1 Model convergence

Out of the 6,400 sampled universes, 1,487 failed to converge. Non-convergence mostly occurred in log-binomial models (n = 1,102), followed by generalised estimating equations (GEE; n = 221) and logistic regression (n = 164). The other three model types (Cox PH, mixed effects logistic, and discrete-time mixed effects) converged successfully in all cases. Across other analytical decisions, non-convergence was distributed approximately equally, suggesting these decisions did not substantially affect model convergence. In total, 4,913 universes yielded valid risk ratio estimates. The number of converged and non-converged universes per decision option can be found in Table 3.

1.2 Overall multiverse results

Across the 4,913 successful specifications, risk ratio estimates ranged from 0.71 to 1.68 (median = 0.95, mean = 0.94, SD = 0.1), with 61.1% of the universes yielding significant results ($p < .05$). Within this sample of the full multiverse, marriage thus appeared both as a protective factor (RR < 1) and a risk factor (RR > 1) for cardiovascular events, depending on the analytical specification. Figure 1 shows the full distribution of the effect estimates across universes, arranged from smallest to largest risk ratio, with significance indicated by colour.

Several patterns emerged in the specification curve (Figure 1). Universes including age as an adjustment variable were clustered around higher RRs (mostly ≥ 1), while universes without age were clustered around lower RRs (all < 1), suggesting a potential effect of age on the relationship between marriage and cardiovascular events. The highest RRs (> 1.2) only occurred in specifications including Cox PH models, although almost none of these estimates reached significance. Log-binomial models, in contrast, yielded only RRs below 1, almost all of which were significant. Furthermore, effect estimates seemed robust to data type (except cross-sectional W1), missing data handling, outcomes (except Age + Ever), and exclusion criteria, with no clear clustering patterns across these decisions.

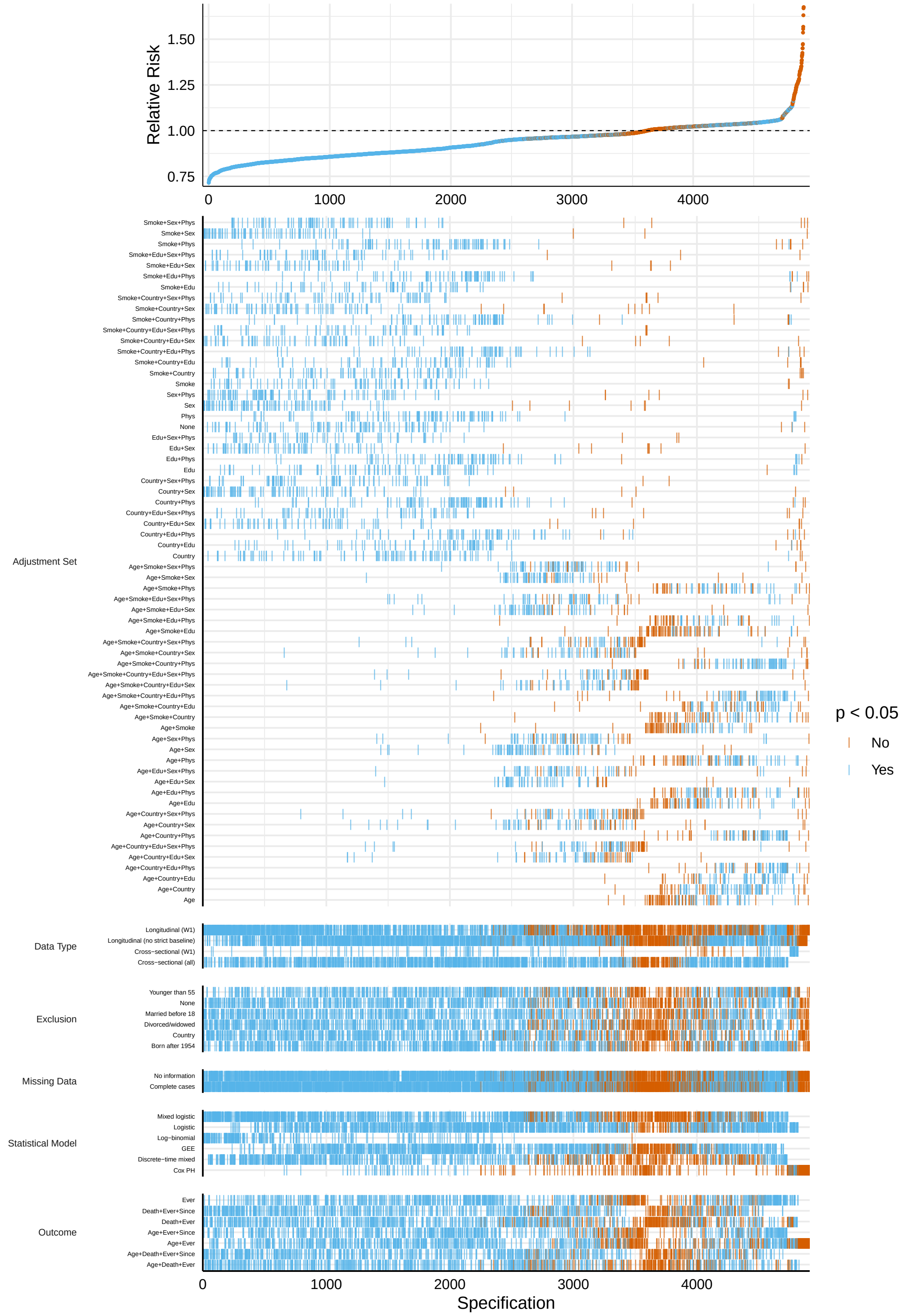

Figure 1: Specification curve multiverse results

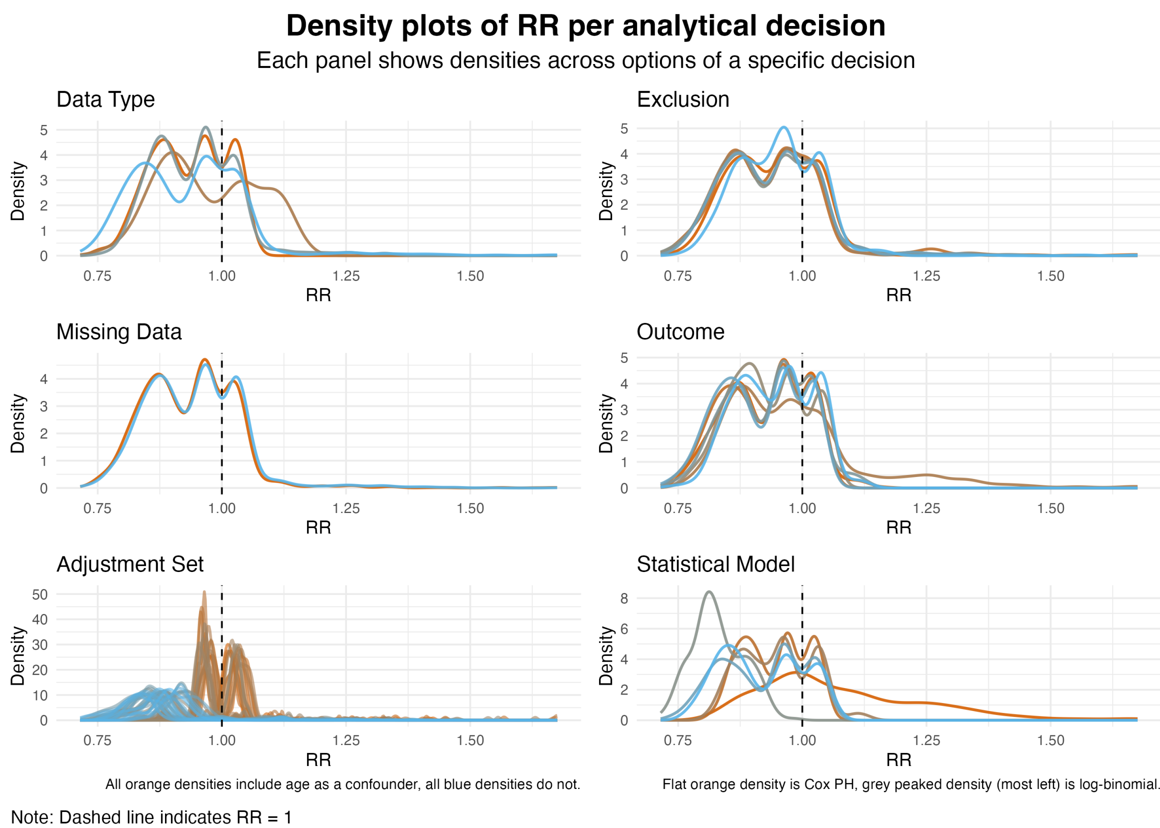

Figure 2: Densities of risk ratios (RR) across decisions and their options.

1.3 Decision sensitivity

To examine the effect size distributions across decision options in more detail, density plots were used (see Figure 2). Data type, missing data, exclusion criteria, and outcome showed largely overlapping RR distributions across their options. However, in case of statistical models and adjustment sets there were clear separations in RR distributions between options. Consistent with the pattern shown in Figure 1, log-binomial models were clustered below RR values of 1, while the density of Cox PH showed a long tail extending above 1. Similarly, specifications including age in the adjustment set showed a clear shift towards higher RRs compared to those without age as a confounder.

Mean RR ranges across options confirmed the patterns found in the specification curve and density plots (Table 3). Statistical model (0.83 - 1.08) and adjustment set (0.83 - 1.05) showed the largest variation in mean RR across options, while other decisions showed smaller ranges: exclusion criteria (0.94 - 0.94), missing data (0.93 - 0.95), data type (0.93 - 0.97), and outcome (0.92 - 0.98).

Table 1: K-sample Anderson-Darling tests for distributional differences across analytical decision options, with higher standardised AD values indicating more sensitive decisions.

Decision	k	T.AD	p-value
Missing Data	2	5.89	0.002
Exclusion Criteria	6	13.13	$< .001$
Data Type	4	50.88	$< .001$
Outcome	7	52.26	$< .001$
Statistical Model	6	307.10	$< .001$
Adjustment Set	64	394.79	$< .001$

Note. k = number of options; T.AD = standardised AD value.

A k-sample Anderson-Darling test (AD test) was used to quantify decision sensitivity, testing whether the RRs from k options within each decision come from the same empirical cumulative distribution function (ECDF; for the ECDF per decision see Figure 4). As shown in Table 1, AD tests for all decisions were highly significant (almost all $p < .001$), indicating distributional differences across options within each decision. The standardised AD statistic (T.AD), which adjusts for number of groups and sample size, was used for between-decision comparisons. Adjustment set and statistical model showed substantially higher T.AD values (307.1 and 394.79, respectively) than other decisions (missing data: 5.89; exclusion: 13.13; outcome: 50.88; data type: 52.26). This indicates that statistical model and adjustment set had the largest impact on the distribution of effect estimates. The number of options per decision did not seem to substantially affect the T.AD values.

Because there were 64 different adjustment sets, for a more detailed assessment of the sensitivity of this decision, Kolmogorov-Smirnov tests were used to compare RR distributions for specifications including versus excluding each adjustment variable. For the ECDFs across adjustment variables, see Figure 5. As can be seen in Table 2, inclusion of all adjustment variables, except smoking, resulted in significantly different distributions than when these variables were excluded ($p < .001$). Age showed the largest effect (0.88), followed by not including any confounders (0.54) and sex (0.45). Physical activity, country, and education showed moderate, but significant, effects ($D_{activity} = 0.16$, $D_{country} = 0.1$, and $D_{education} = 0.08$). This confirms that the adjustment set sensitivity seems to be driven primarily by age, sex and the decision not to include confounders at all.

Table 2: Kolmogorov–Smirnov test for distributional differences across adjustment set options, when including or excluding specific adjustment variables. D: absolute max distance between the ECDFs of the two samples. Values closer to 0 the indicate higher likelihood that the two samples were drawn from the same distribution.

Adjustment variable	Included	Excluded	D	p-value
Smoking	2332	2581	0.03	0.341
Education	2214	2699	0.08	$< .001$
Country	2436	2477	0.10	$< .001$
Physical activity	2431	2482	0.16	$< .001$
Sex	2409	2504	0.45	$< .001$
None	100	4813	0.54	$< .001$
Age	2330	2583	0.88	$< .001$

Note. D = KS test statistic (range: 0 - 1), smaller numbers indicate smaller distributional differences.

The adjustment set sensitivity driven by age, sex and none is also visible in the RR density distributions (Figure 3). Including age as an adjustment variable shifted the RR distribution to the right (higher RRs), while including sex shifted it to the left (lower RRs). Specifications without any confounders also showed lower RRs. Other adjustment variables did not result in pronounced differences between RR distributions.

In the underlying data, married individuals were on average 4 years younger than unmarried individuals (65 versus 69 years old). Additionally, the distribution of sex differed between groups: 51% of married individuals were female, compared to 69% in the unmarried group.

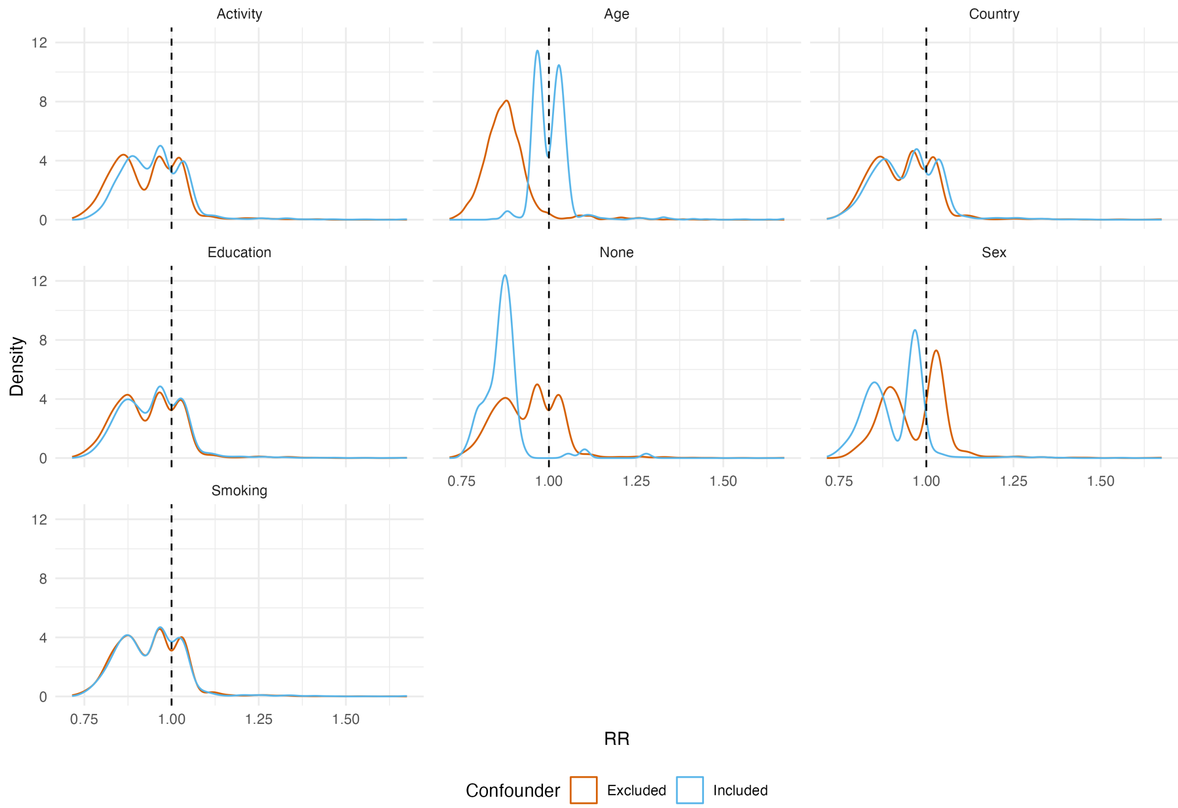

Figure 3: Densities of risk ratios (RR) across inclusion/exclusion of different adjustment variables.

2 Appendix

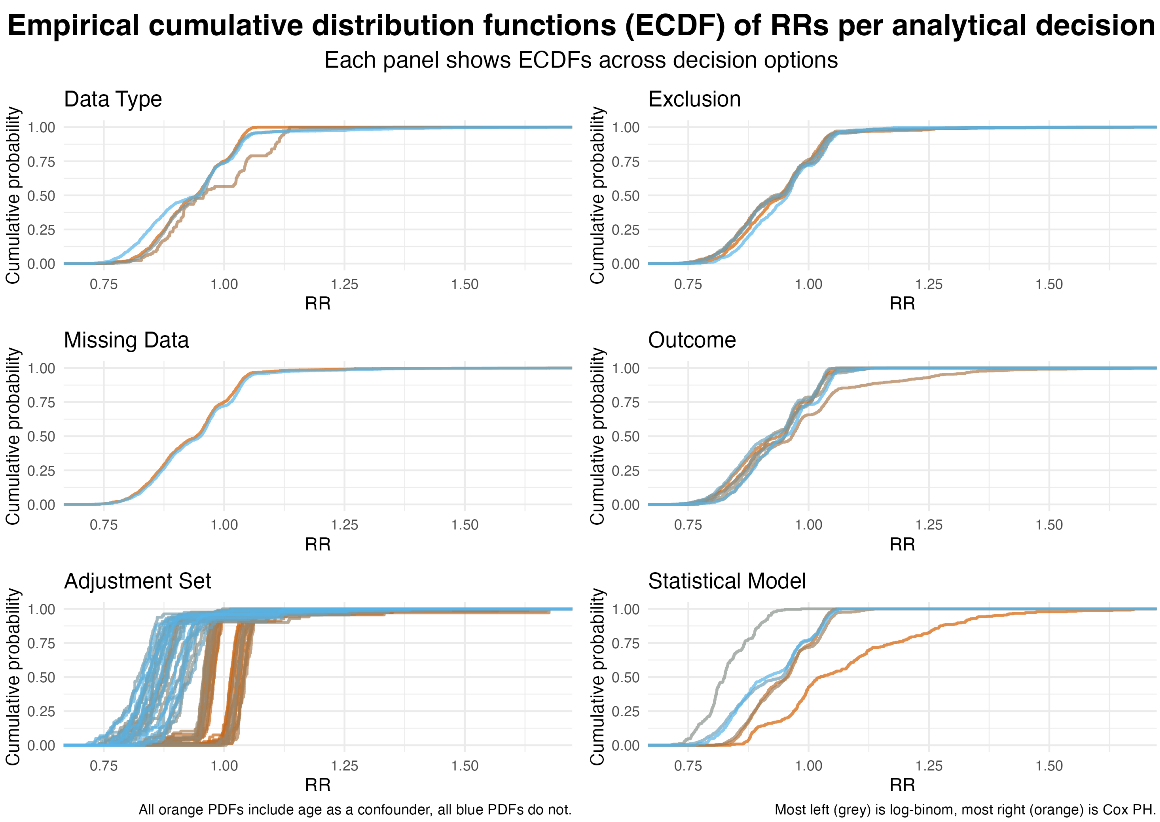

Figure 4: Empirical cumulative distribution functions of risk ratios (RR) across decisions and their options.

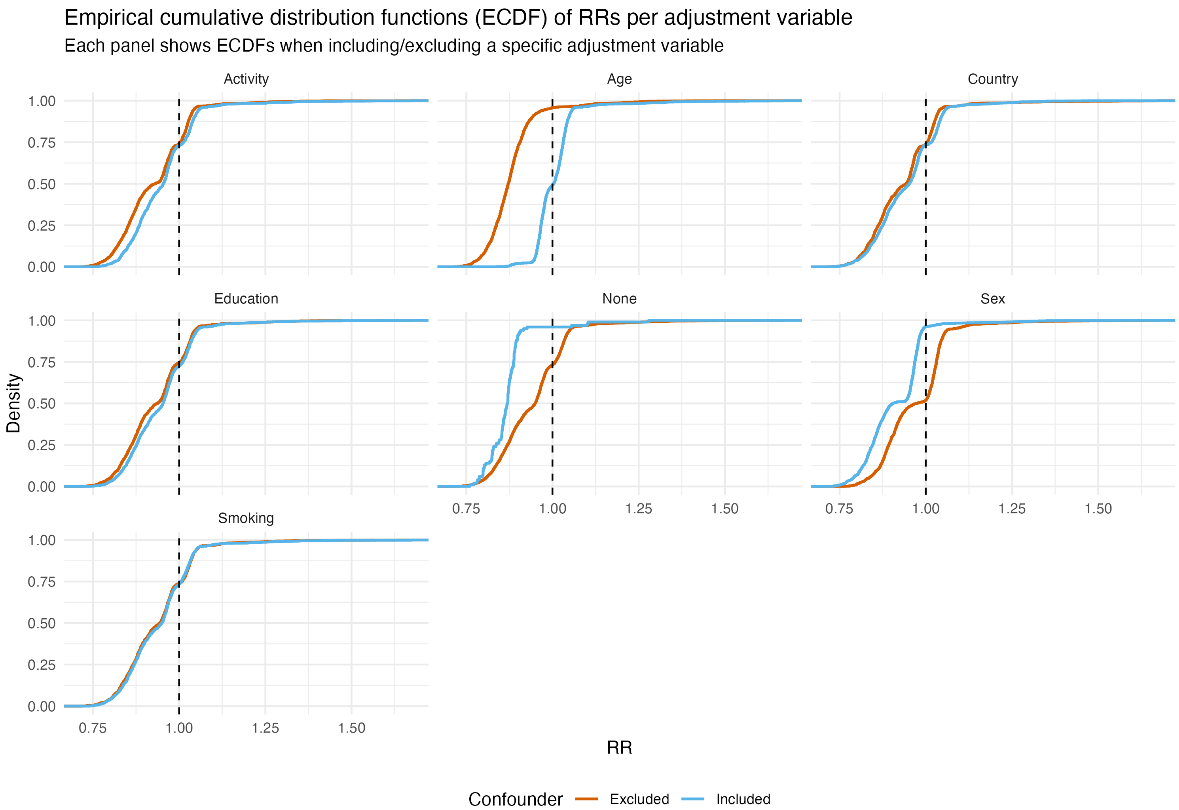

Figure 5: Empirical cumulative distribution functions of risk ratios (RR) across inclusion/exclusion of different adjustment variables.

Table 3: Number of universes including specific decision options, non-convergence, RRs, their range and percentage of significant RRs across decisions and their options.

Decision	Option	Succesful Universes	Non-convergence	Mean RR	RR Range	% Sig
Statistical Model	Log-binomial	232	1102	0.83	[0.71, 0.98]	100%
Statistical Model	Mixed effects logistic regression	1278	0	0.92	[0.73, 1.07]	77%
Statistical Model	Discrete-time mixed effects	832	0	0.92	[0.76, 1.07]	74%
Statistical Model	GEE	1052	221	0.95	[0.8, 1.06]	85%
Statistical Model	Logistic regression	1188	164	0.95	[0.8, 1.14]	93%
Statistical Model	Cox PH	331	0	1.08	[0.84, 1.68]	24%
Data Type	Cross-sectional (all waves)	1237	403	0.93	[0.74, 1.07]	91%
Data Type	Longitudinal (wave 1)	1669	563	0.93	[0.71, 1.68]	69%
Data Type	Longitudinal (no strict baseline)	1869	411	0.95	[0.75, 1.35]	82%
Data Type	Cross-sectional (wave 1)	138	110	0.97	[0.77, 1.14]	86%
Missing Data	Complete cases	2584	637	0.94	[0.72, 1.68]	80%
Missing Data	No information	2329	850	0.94	[0.71, 1.68]	80%
Exclusion Criteria	Country	833	261	0.93	[0.73, 1.54]	76%
Exclusion Criteria	Divorced/widowed	819	277	0.93	[0.73, 1.47]	79%
Exclusion Criteria	Married before 18	778	248	0.94	[0.71, 1.45]	82%
Exclusion Criteria	None	804	263	0.94	[0.73, 1.56]	80%
Exclusion Criteria	Born after 1954	886	209	0.95	[0.74, 1.68]	83%

(continued)

Decision	Option	Succesful Universes	Non-convergence	Mean RR	RR Range	% Sig
Exclusion Criteria	Younger than 55	793	229	0.95	[0.76, 1.67]	79%
Outcome	Age event + Death + Event ever + Since	644	193	0.92	[0.73, 1.06]	84%
Outcome	Death + Event ever + Since	656	182	0.92	[0.71, 1.06]	84%
Outcome	Age event + Death + Event ever	707	177	0.93	[0.75, 1.16]	76%
Outcome	Age event + Event ever + Since	617	216	0.93	[0.74, 1.07]	86%
Outcome	Death + Event ever	761	187	0.93	[0.72, 1.13]	76%
Outcome	Event ever	695	266	0.95	[0.73, 1.14]	83%
Outcome	Age event + Event ever	833	266	0.98	[0.77, 1.68]	72%
Adjustment Set	Smoke + Sex	81	19	0.83	[0.74, 1.35]	97%
Adjustment Set	Country + Sex	100	0	0.83	[0.73, 1.12]	95%
Adjustment Set	Sex	100	0	0.83	[0.71, 1.22]	93%
Adjustment Set	Smoke + Edu + Sex	65	35	0.84	[0.75, 1.01]	93%
Adjustment Set	Smoke + Country + Edu + Sex	74	26	0.85	[0.74, 1.16]	94%
Adjustment Set	Smoke + Country + Sex	85	15	0.85	[0.75, 1.03]	90%
Adjustment Set	Edu + Sex	75	25	0.85	[0.76, 1.31]	90%
Adjustment Set	Sex + Phys	100	0	0.85	[0.76, 1.39]	94%
Adjustment Set	Smoke	86	14	0.86	[0.77, 1.08]	98%
Adjustment Set	Smoke + Edu + Sex + Phys	66	34	0.86	[0.77, 1.2]	96%
Adjustment Set	Country + Edu + Sex	73	27	0.86	[0.75, 1.22]	91%
Adjustment Set	Country + Sex + Phys	79	21	0.86	[0.77, 1.01]	97%
Adjustment Set	Edu + Sex + Phys	68	32	0.86	[0.8, 1.02]	96%
Adjustment Set	Smoke + Country + Edu + Sex + Phys	63	37	0.87	[0.78, 1]	93%
Adjustment Set	Smoke + Country + Sex + Phys	72	28	0.87	[0.77, 1.21]	92%
Adjustment Set	Smoke + Sex + Phys	84	16	0.87	[0.8, 1.38]	94%
Adjustment Set	None	100	0	0.87	[0.76, 1.28]	97%
Adjustment Set	Smoke + Country	87	13	0.88	[0.78, 1.26]	96%
Adjustment Set	Country	100	0	0.88	[0.76, 1.24]	98%
Adjustment Set	Country + Edu + Sex + Phys	73	27	0.89	[0.79, 1.25]	91%
Adjustment Set	Edu	78	22	0.89	[0.79, 1.13]	99%
Adjustment Set	Smoke + Country + Edu	66	34	0.90	[0.79, 1.21]	93%
Adjustment Set	Smoke + Edu	67	33	0.90	[0.79, 1.41]	95%
Adjustment Set	Phys	100	0	0.90	[0.81, 1.12]	100%
Adjustment Set	Country + Edu	75	25	0.91	[0.81, 1.21]	93%
Adjustment Set	Smoke + Country + Phys	84	16	0.92	[0.82, 1.1]	94%
Adjustment Set	Smoke + Phys	87	13	0.92	[0.81, 1.25]	93%
Adjustment Set	Country + Phys	100	0	0.92	[0.82, 1.31]	96%
Adjustment Set	Edu + Phys	79	21	0.93	[0.83, 1.24]	98%
Adjustment Set	Smoke + Country + Edu + Phys	69	31	0.94	[0.83, 1.27]	95%
Adjustment Set	Smoke + Edu + Phys	75	25	0.94	[0.84, 1.47]	95%
Adjustment Set	Country + Edu + Phys	72	28	0.94	[0.83, 1.27]	92%
Adjustment Set	Age + Smoke + Sex	74	26	0.96	[0.87, 1.24]	82%
Adjustment Set	Age + Country + Edu + Sex	69	31	0.96	[0.87, 1.02]	63%
Adjustment Set	Age + Edu + Sex	60	40	0.96	[0.88, 1.09]	80%
Adjustment Set	Age + Sex	82	18	0.96	[0.88, 1.03]	88%
Adjustment Set	Age + Smoke + Country + Edu + Sex	62	38	0.97	[0.84, 1.3]	69%
Adjustment Set	Age + Smoke + Edu + Sex	68	32	0.97	[0.89, 1.41]	79%
Adjustment Set	Age + Smoke + Sex + Phys	79	21	0.97	[0.9, 1.33]	77%
Adjustment Set	Age + Country + Sex	77	23	0.97	[0.87, 1.28]	68%
Adjustment Set	Age + Sex + Phys	79	21	0.97	[0.88, 1.56]	78%
Adjustment Set	Age + Smoke + Country + Edu + Sex + Phys	61	39	0.98	[0.88, 1.26]	54%
Adjustment Set	Age + Smoke + Country + Sex	75	25	0.98	[0.84, 1.35]	65%
Adjustment Set	Age + Smoke + Edu + Sex + Phys	62	38	0.98	[0.88, 1.57]	80%
Adjustment Set	Age + Country + Edu + Sex + Phys	80	20	0.98	[0.87, 1.33]	57%
Adjustment Set	Age + Country + Sex + Phys	86	14	0.98	[0.85, 1.33]	57%
Adjustment Set	Age + Edu + Sex + Phys	65	35	0.98	[0.88, 1.33]	76%
Adjustment Set	Age + Smoke + Country + Sex + Phys	72	28	0.99	[0.87, 1.32]	43%
Adjustment Set	Age + Smoke	75	25	1.01	[0.92, 1.04]	37%
Adjustment Set	Age	87	13	1.02	[0.98, 1.42]	33%
Adjustment Set	Age + Smoke + Edu	80	20	1.02	[0.96, 1.13]	27%
Adjustment Set	Age + Smoke + Edu + Phys	61	39	1.03	[0.94, 1.67]	51%
Adjustment Set	Age + Smoke + Country	73	27	1.04	[0.95, 1.42]	44%
Adjustment Set	Age + Smoke + Country + Edu	67	33	1.04	[0.96, 1.2]	60%
Adjustment Set	Age + Smoke + Country + Edu + Phys	66	34	1.04	[0.94, 1.45]	77%

(continued)

Decision	Option	Succesful Universes	Non-convergence	Mean RR	RR Range	% Sig
Adjustment Set	Age + Country	87	13	1.04	[0.98, 1.28]	74%
Adjustment Set	Age + Edu	71	29	1.04	[0.99, 1.63]	48%
Adjustment Set	Age + Phys	85	15	1.04	[0.98, 1.34]	60%
Adjustment Set	Age + Smoke + Country + Phys	78	22	1.05	[0.98, 1.4]	82%
Adjustment Set	Age + Smoke + Phys	68	32	1.05	[0.97, 1.68]	58%
Adjustment Set	Age + Country + Edu	63	37	1.05	[0.98, 1.37]	75%
Adjustment Set	Age + Country + Edu + Phys	71	29	1.05	[1, 1.47]	78%
Adjustment Set	Age + Country + Phys	77	23	1.05	[0.99, 1.45]	70%
Adjustment Set	Age + Edu + Phys	70	30	1.05	[1, 1.54]	59%